

Lake Michigan Shoreline Delineation Documentation

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Tools Used: ArcGIS Pro (v3.1), Digital Shoreline Analysis System (DSAS v6.0.170)

1. Introduction

This documentation outlines the digitization and analysis process for delineating the Lake Michigan shoreline using a combination of ArcGIS Pro and DSAS. The shoreline data for 2012 and 2020 were extracted, cleaned, and prepared for analysis to quantify shoreline change across Illinois and Indiana.

2. Extracting the DEM and Preliminary Shoreline

We began by extracting the LiDAR-derived Digital Elevation Models (DEMs) from NOAA for the years 2012 and 2020. The DEMs were selected with the vertical datum set to IGLD85.

Tip: Use the “Bulk Download” option and retrieve the LiDAR DEMs not the points as they don’t have the vertical datum IGLD85.

Following the shoreline extraction guide [Manual for retrieving specific contour lines from TopobathyTiff.docx 1.pdf](#), we derived contours using a base elevation of **177.2 m**, which corresponds to the agreed Ordinary High-Water Mark (OHWM). This differs from the reference manual’s 177.45 m.

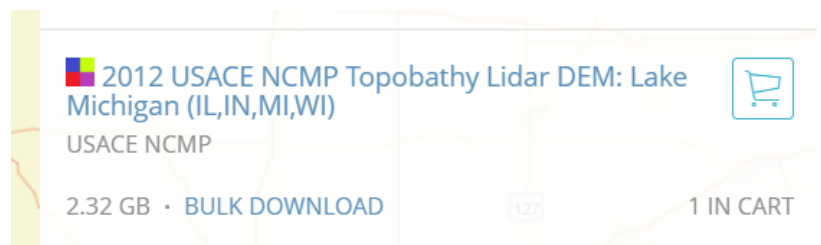


Fig. 1. Reference for the downloaded tile for the year 2012

Because DEMs are split into tiles, we either:

- **Mosaicked** them using the **Mosaic to New Raster** tool, or
- **Individually extracted shorelines** from each tile and **merged the shapefiles** afterward.

We chose the latter to maintain higher spatial accuracy.

3. Data Cleanup

Data cleaning was a critical step, especially due to missing DEM sections in areas like Chicago.



Fig.2. Missing DEM data for Chicago, the red is the extracted shoreline of the year 2012

To compensate for this and get a stable result for the analysis, the shorelines of both the year 2012 and 2020 were made of similar coverage.

Handling Gaps and Artifacts

- **Visual inspection** using Google Earth (for the respective years) guided manual gap filling.
- Emphasis was placed on **natural features and beach areas**.
- **Irrelevant structures** (e.g., incomplete harbours sections) were deleted using the **Edit tool**.

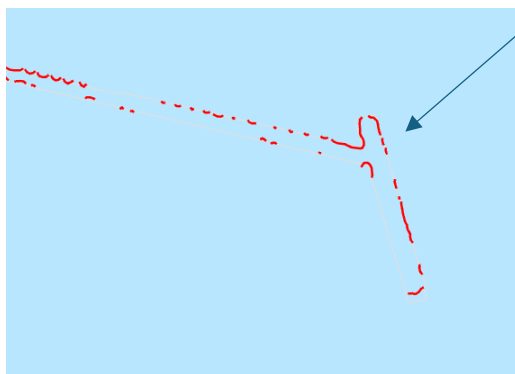
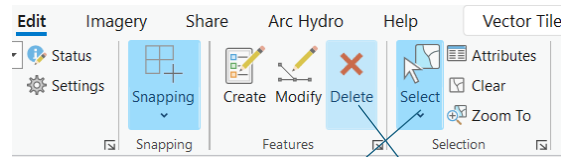


Fig.3 Before

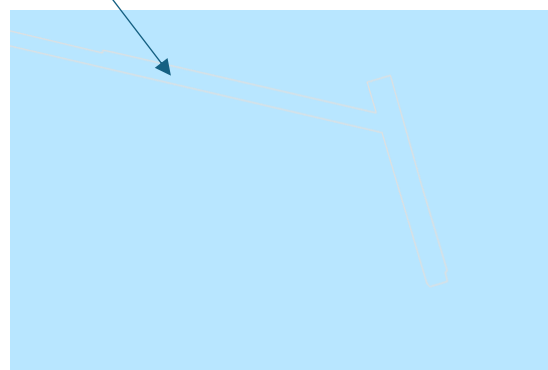


Fig.4 After

⚠ Caution: Always verify your selection before deleting. Use zoom-in/out to avoid deleting large sections unintentionally. Avoid saving immediately so mistakes can be reversed.

Connecting Line Gaps

- Enable **Snapping** under the **Edit tab**.

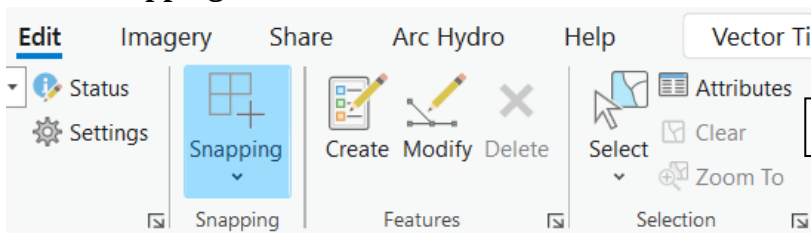


Fig.5 The edit tab

- Use **Edit Vertices** to manually drag endpoints into place.

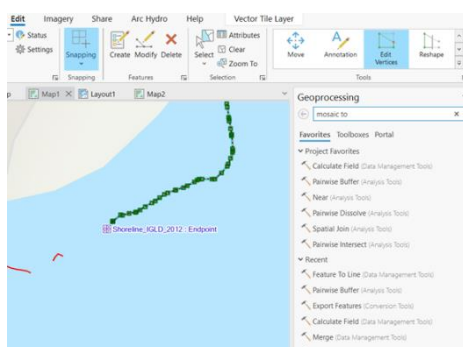


Fig.6 Toggling the edit vertices option

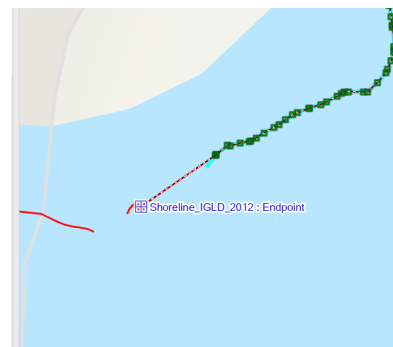


Fig.7 Manually dragging the endpoints

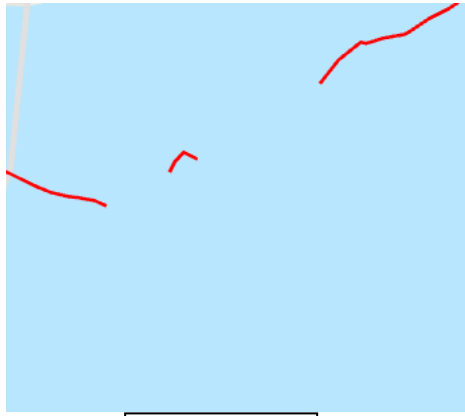


Fig.8 Before

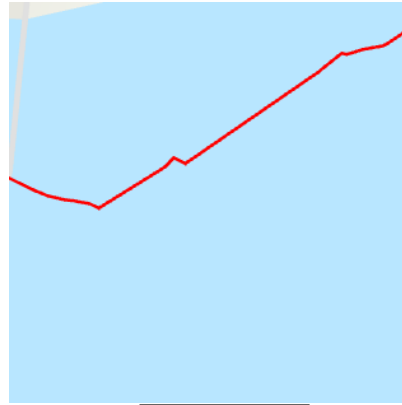


Fig.9 After

- Always refer to historical basemaps to ensure trace accuracy.

After cleaning, all shoreline shapefiles were merged using the **Merge** tool to preserve attributes.

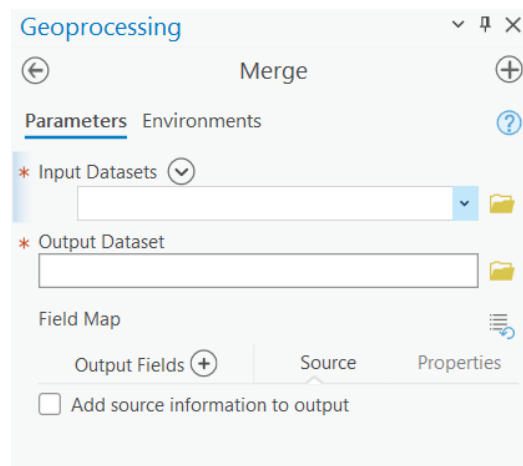


Fig.10 Merge Tool

4. Creating the Baseline

The **baseline** was generated using the **Buffer** tool:

- Created a **100 m planar buffer** from the 2012 shoreline.

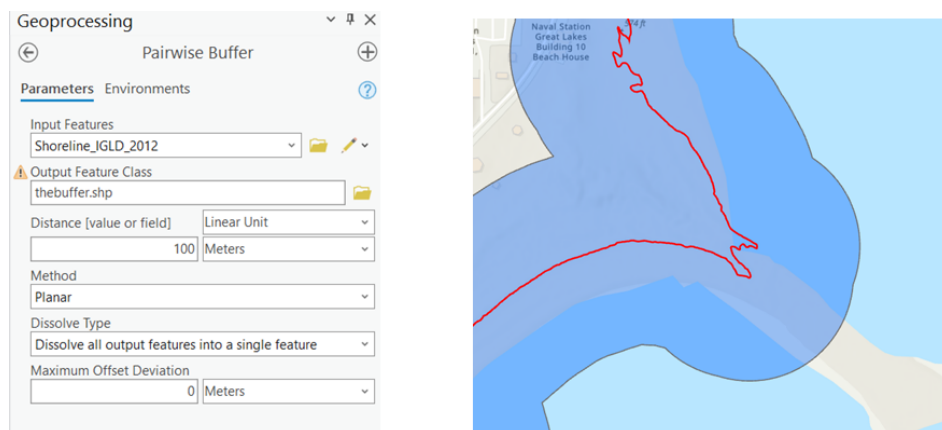


Fig.11 The Buffer tool and the created buffer

- Converted the buffer polygon to line using the **Feature to Line** tool.

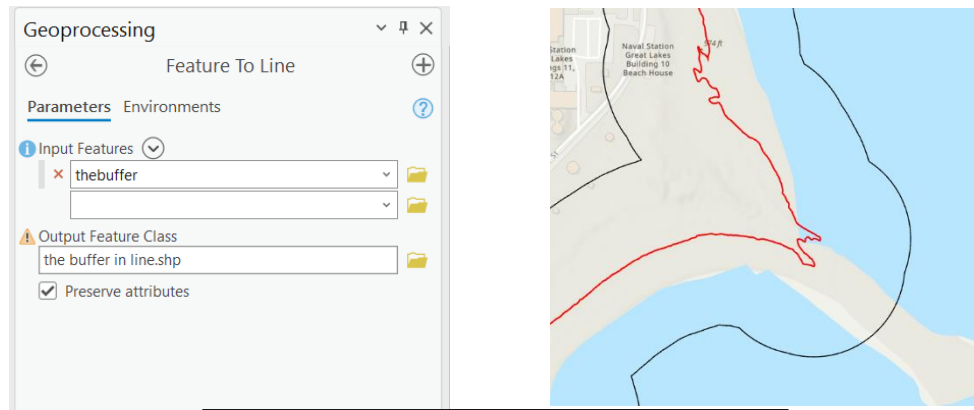


Fig.12 The feature to line tool and the result

- Removed offshore segments using the **Split** tool and manual editing.

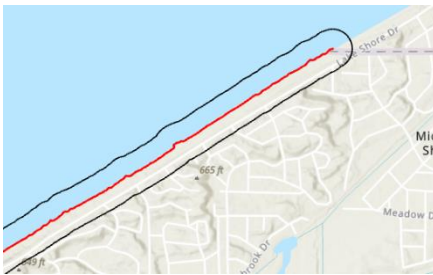


Fig.13 The selection

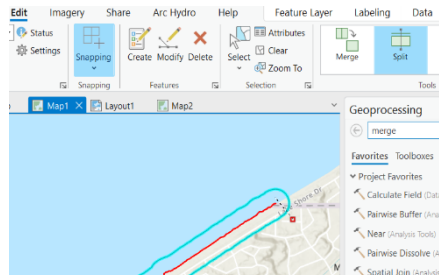


Fig.14 The splitting process

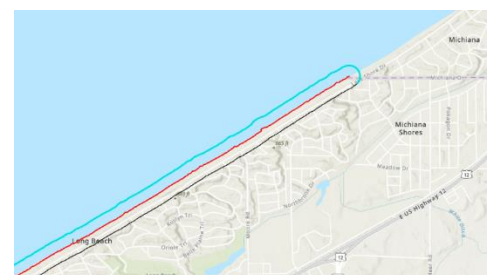


Fig.15 The separated result

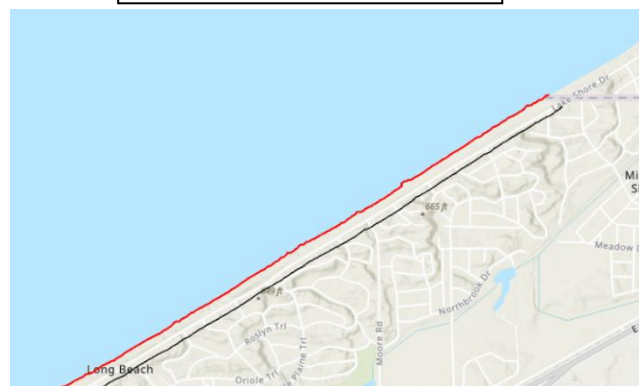


Fig.16 The resulting baseline

Note: Ensure the baseline direction flows **West to East**. If needed, use the **Flip Line** tool. This change won't be visible in ArcGIS Pro but will be recognized in DSAS.

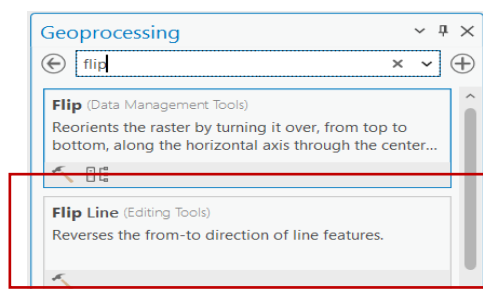


Fig.17 The flip line tool

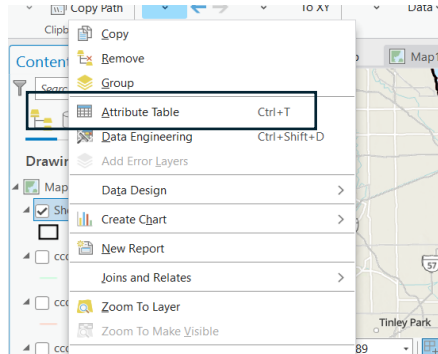
Finally, merged all segments into a single baseline shapefile.

5. Adding Flight Dates to Shoreline Features

Flight dates were obtained from the ArcGIS layer packages ([The_USACE_2020_Coverage.lpkx](#) and [The_USACE_2012_Coverage.lpkx](#)). These require **ArcGIS Pro v3.4 or higher** to open.

Steps:

- Open the **Attribute Table**.



- Use **Add Field** to include a “Date” field.

A screenshot of the 'Add Field' dialog box in ArcGIS Pro. The 'Add' button is highlighted with a blue box. The dialog shows a table with columns: FID, Shape *, FID_thebuf, and Id. The table contains 12 rows of data, with the last row (FID 12) highlighted in blue. The 'Add' button is located at the top left of the dialog box.

FID	Shape *	FID_thebuf	Id
5	4	Polyline	0 0
6	5	Polyline	0 0
7	6	Polyline	0 0
8	7	Polyline	0 0
9	8	Polyline	0 0
10	9	Polyline	0 0
11	10	Polyline	0 0
12	11	Polyline	0 0

- Add field name, data type and other relevant attributes.

Current Layer: Shoreline_IGLD_2012 (Map1)

	Visible	Read Only	Field Name	Alias	Data Type	Allow NULL	Highlight	Number Format	Default	Precision	Scale	Length
	☒	☒	FID	FID	Object ID	☐	☐	Numeric		0	0	
	☒	☐	Shape	Shape	Geometry	☐	☐			0	0	
	☒	☐	Id	Id	Long	☐	☐	Numeric		10	0	
	☒	☐	Contour	Contour	Double	☐	☐	Numeric		0	0	
	☒	☐	Date	Date	Date	☐	☐			0	0	
	☒	☐	Field		Long	☐	☐					

- Populate flight dates accordingly

Shoreline_IJGLD_2012					
Shoreline					
Field: Add Calculate Selection:					
	FID	Shape *	Id	Contour	Date
1	0	Polyline	2	177.2	1/1/2012
2	1	Polyline	41	177.2	1/1/2012
3	2	Polyline	42	177.2	1/1/2012
4	3	Polyline	43	177.2	1/1/2012
5	4	Polyline	44	177.2	1/1/2012
6	5	Polyline	45	177.2	1/1/2012
7	6	Polyline	46	177.2	1/1/2012
8	7	Polyline	47	177.2	1/1/2012

Even after merging, the shapefile may appear fragmented. This doesn't impact DSAS performance.

6. Exporting to GeoJSON for DSAS tool

To run the shoreline change analysis DSAS will require a merged shapefile of both the 2012 and 2020 shorelines as the main shoreline, else there is a slight chance that it may consider the baseline as the second shoreline and thereby ending up in wrong analysis and calculations, so make sure to merge both the cleaned shapefile of both the years as one single shapefile as a Shoreline shapefile for DSAS. We only need to add the dates to the Shoreline shapefiles and not to the baseline.

DSAS requires input files in **GeoJSON format**. Use the **Features to JSON** tool in ArcGIS to convert shapefiles.

Geoprocessing

Features To JSON

Parameters Environments

Input Features

Shoreline

Output JSON

D:\PURDUE\SEM 4\Independent Study - Dr.Troy\GIS

☐ Formatted JSON
☒ Output to GeoJSON
☐ Project to WGS_1984
☒ Use field aliases

If the tool fails on your system, open the shapefiles in **QGIS** and export as GeoJSON.

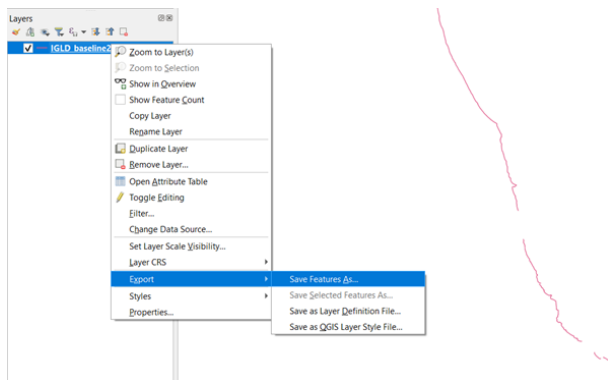


Fig.18 Step 1

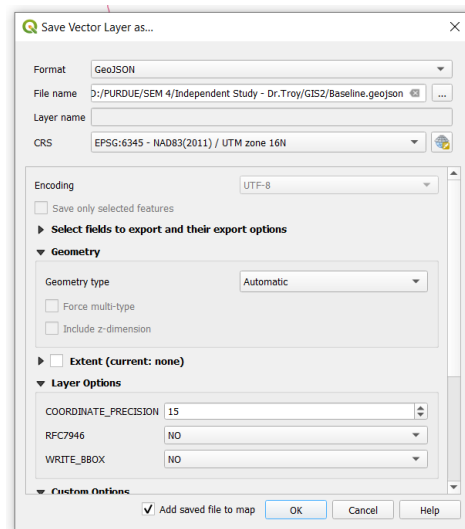


Fig.19 Step 2

⚠ After any edit, inspect the **FID** and other ID fields. Ensure values are in cardinal (sequential) order to avoid attribute errors in DSAS.

8. Running DSAS Analysis

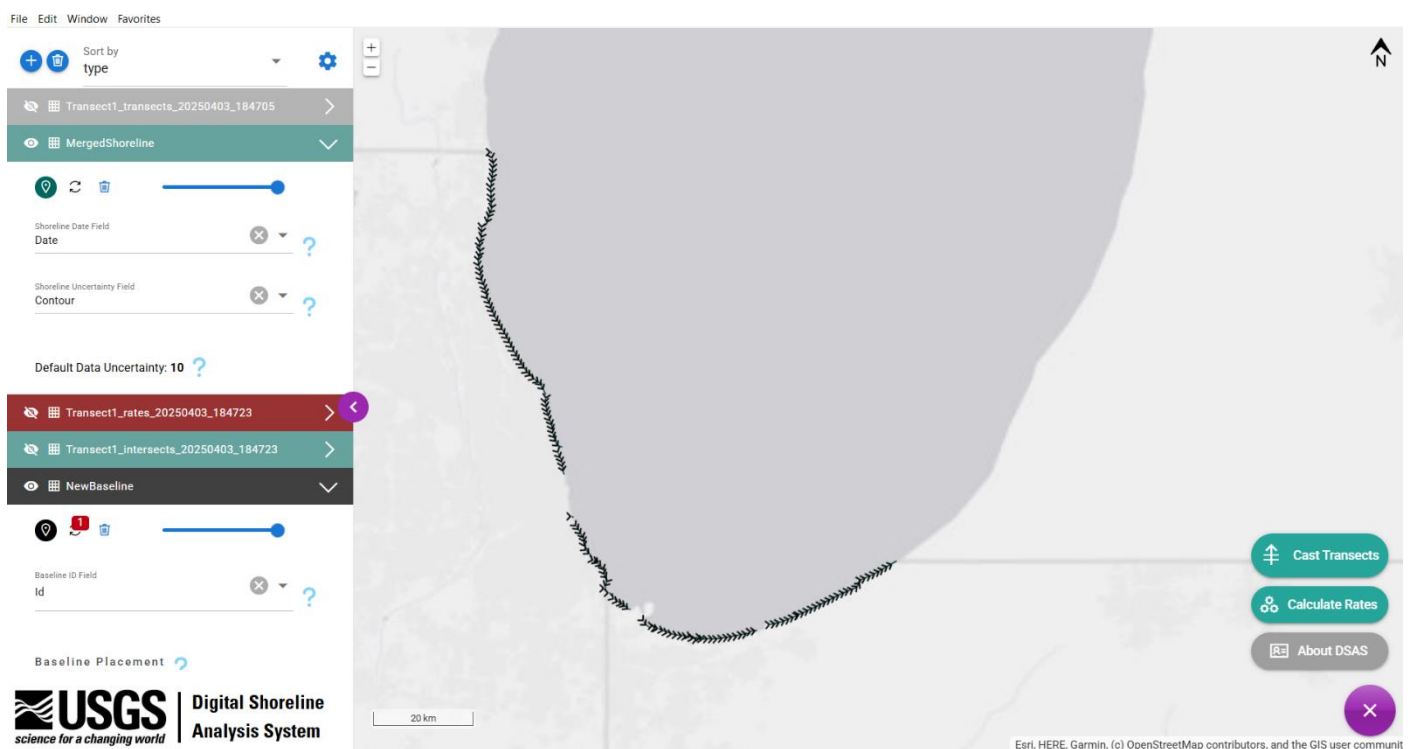


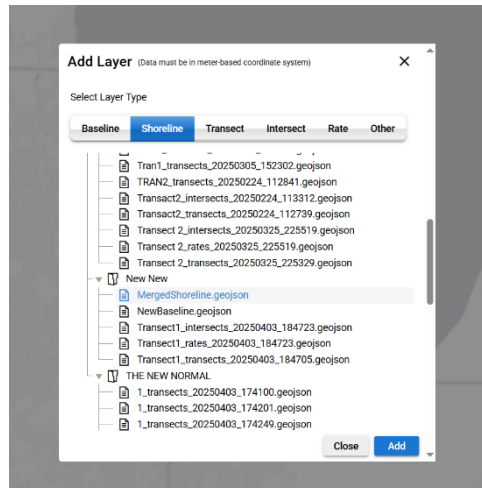
Fig.20 A typical DSAS window setup

Before beginning:

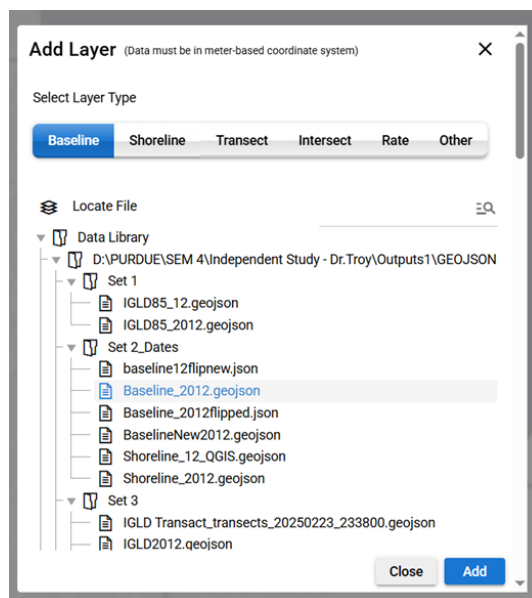
- Create a **dedicated folder** for DSAS outputs.
- Add the merged **shoreline** and **baseline** GeoJSON files.

DSAS Setup:

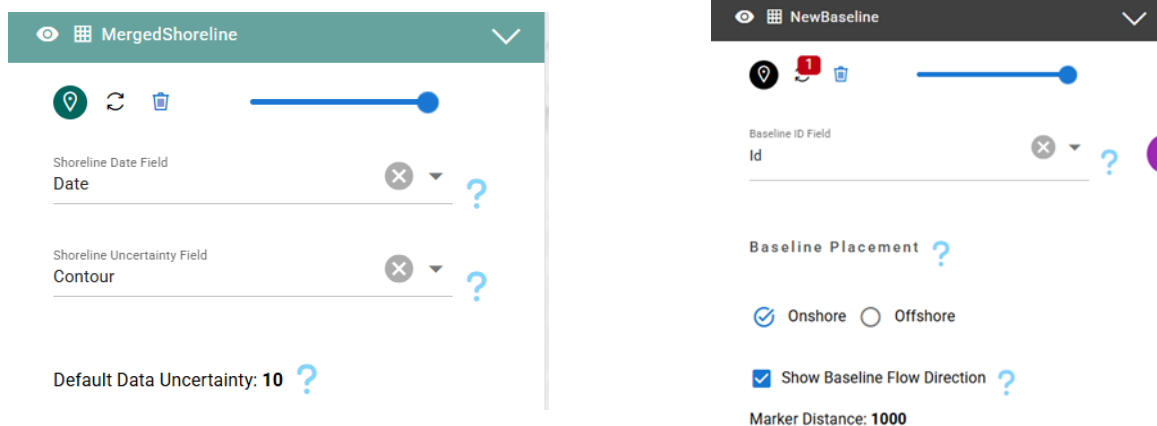
1. Click **Add Layer** → add Shoreline GeoJSON.



2. Toggle **Baseline** → add Baseline GeoJSON.



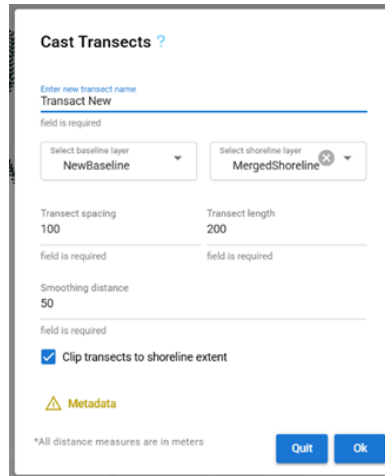
Once you are done adding both the shoreline and baseline, make sure these options are all toggled, and it should be looking like this:



3. Confirm fields for dates and IDs are selectable. If not, check attribute tables.

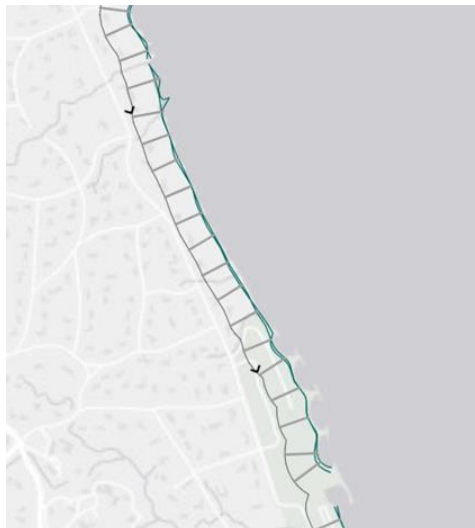
Generating Transects:

- Use **Cast Transect** with your desired settings. Make sure that you clip the transects to shoreline.



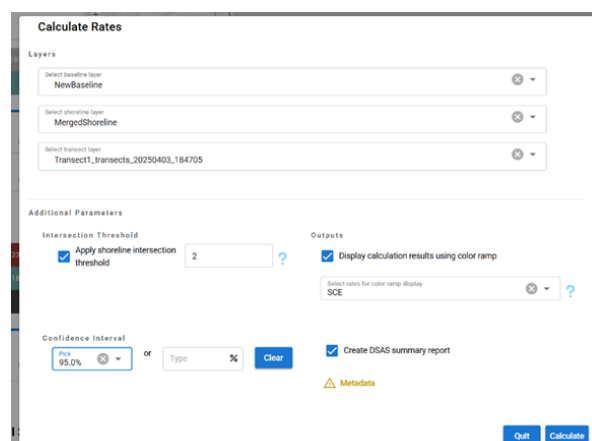
The 'Cast Transects' dialog box is shown. It has a title bar 'Cast Transects ?'. Below the title bar is a section 'Enter new transect name' with a text input field 'Transect New' and a 'field is required' error message. There are two dropdown menus: 'Select baseline layer' with 'NewBaseline' selected, and 'Select shoreline layer' with 'MergedShoreline' selected. Below these are two input fields: 'Transect spacing' with '100' and 'field is required' error, and 'Transect length' with '200' and 'field is required' error. There is a 'Smoothing distance' input field with '50' and 'field is required' error. A checkbox 'Clip transects to shoreline extent' is checked. There is a 'Metadata' section with a warning icon. At the bottom, there is a note '*All distance measures are in meters' and 'Quit' and 'Ok' buttons.

- Verify transects have been generated visually.



Calculating Rates:

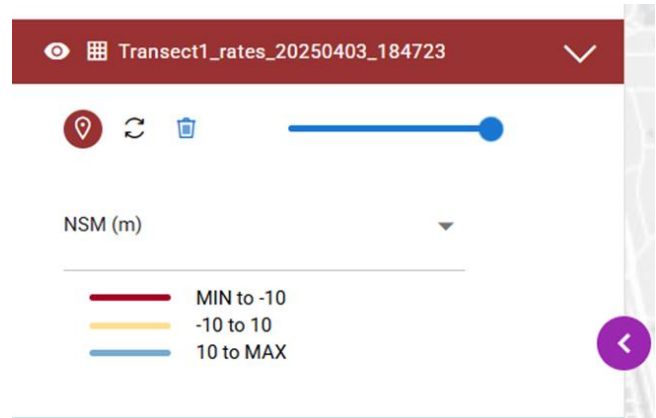
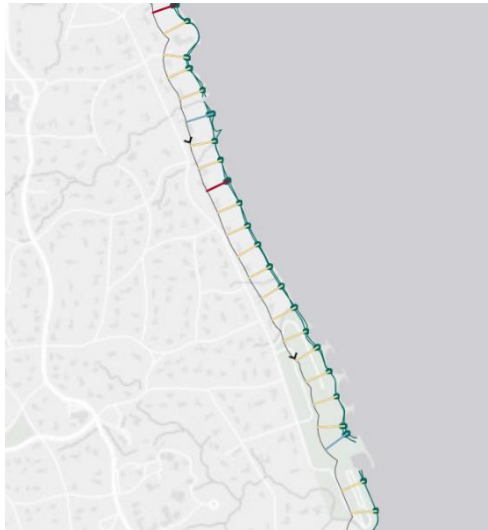
- Click **Calculate Rate**.
- Configure rate types and output options.



The 'Calculate Rates' dialog box is shown. It has a title bar 'Calculate Rates'. Below the title bar is a 'Layers' section with three dropdown menus: 'Select baseline layer' with 'NewBaseline', 'Select shoreline layer' with 'MergedShoreline', and 'Select transect layer' with 'Transect1_transects_20250403_184705'. Below the layers section is an 'Additional Parameters' section. It has an 'Intersection Threshold' section with a checkbox 'Apply shoreline intersection threshold' checked and a value '2'. There is a 'Confidence Interval' section with a dropdown '95.0%' and a 'Type' dropdown with a 'Clear' button. There is an 'Outputs' section with a checkbox 'Display calculation results using color ramp' checked and a dropdown 'SCE'. There is a checkbox 'Create DSAS summary report' and a 'Metadata' section with a warning icon. At the bottom, there are 'Quit' and 'Calculate' buttons.

NSM calculation was used, its displayed as SCE here, kindly ignore that.

- Upon success, DSAS will generate rate layers and legends.



Exporting Summary:

- A text-based **DSAS Summary Report** is auto-generated and can be found in the DSAS folder.
- It contains statistics and metrics based on transects and rate calculations.

```

I:\transsects_20250403_183011.xml 4/3/2023 6:30 PM MICROSOFT EDGE H1... 11 KB
DSAS_summary_report_20250403_18.txt - Notepad
File Edit Format View Help
minimum distance: 0.01
minimum distance transect ID: 532

DISTANCE: NSM (Net Shoreline Movement, m)

NSM OVERALL AVERAGES:
total number of transects: 1375
average distance: -6.20
number of transects with negative distance: 929
percent of all transects that have a negative distance: 67.56%
maximum negative distance: -96.92
maximum negative distance transect ID: 936
average of all negative distances: -12.31
number of transects with positive distance: 446
percent of all transects that have a positive distance: 32.44%
maximum positive distance: 92.63
maximum positive distance transect ID: 497
average of all positive distances: 6.53

```

9. Conclusion

This workflow outlines the step-by-step procedure for digitizing, cleaning, and analyzing the Lake Michigan shoreline change using ArcGIS Pro and DSAS. The methodology ensures reproducible, quality-assured data for coastal engineering and change assessment.